



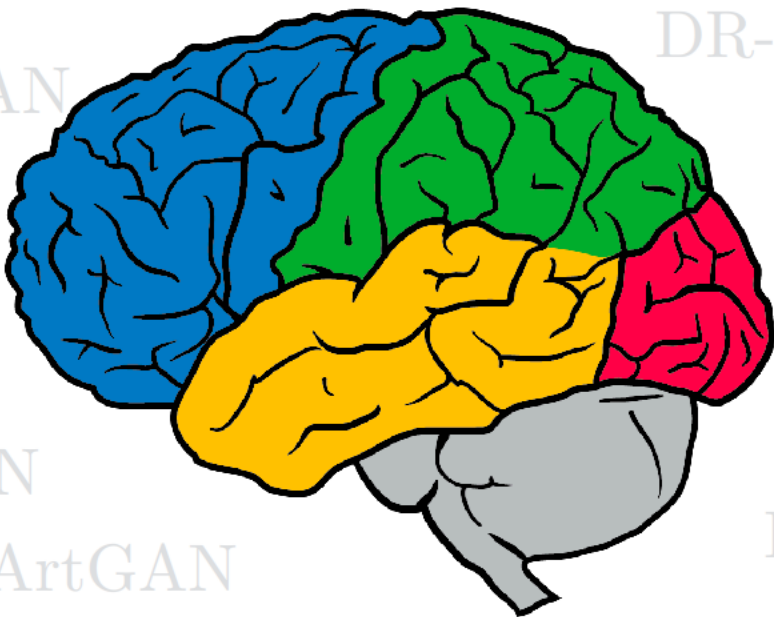
MedGAN ID-CGAN CoGAN LR-GAN CGAN IcGAN
b-GAN LS-GAN AffGAN LAPGAN DiscoGAN MPM-GAN AdaGAN
LSGAN InfoGAN CatGAN AMGAN iGAN IAN

Generative Adversarial Networks

McGAN Ian Goodfellow, Staff Research Scientist, Google Brain MIX+GAN

MGAN alpha-GAN
FF-GAN
2023

C-VAE-GAN C-RNN-GAN
MAGAN 3D-GAN CCGAN
GAWWN DualGAN
Bayesian GAN
EBGAN WGAN-GP
ALI Context-RNN-GAN
MARTA-GAN f-GAN ArtGAN
DR-GAN AC-GAN DCGAN
GoGAN BS-GAN
BiGAN
GP-GAN
AnoGAN
MAD-GAN DTN
BEGAN MalGAN AL-CGAN



Which Face Is Fake?



1



2



3



4

Which Face Is Fake?

Imagined by a GAN (generative adversarial network)
StyleGAN2 (Dec 2019) - Karras et al. and Nvidia



FAKE



FAKE



FAKE



FAKE

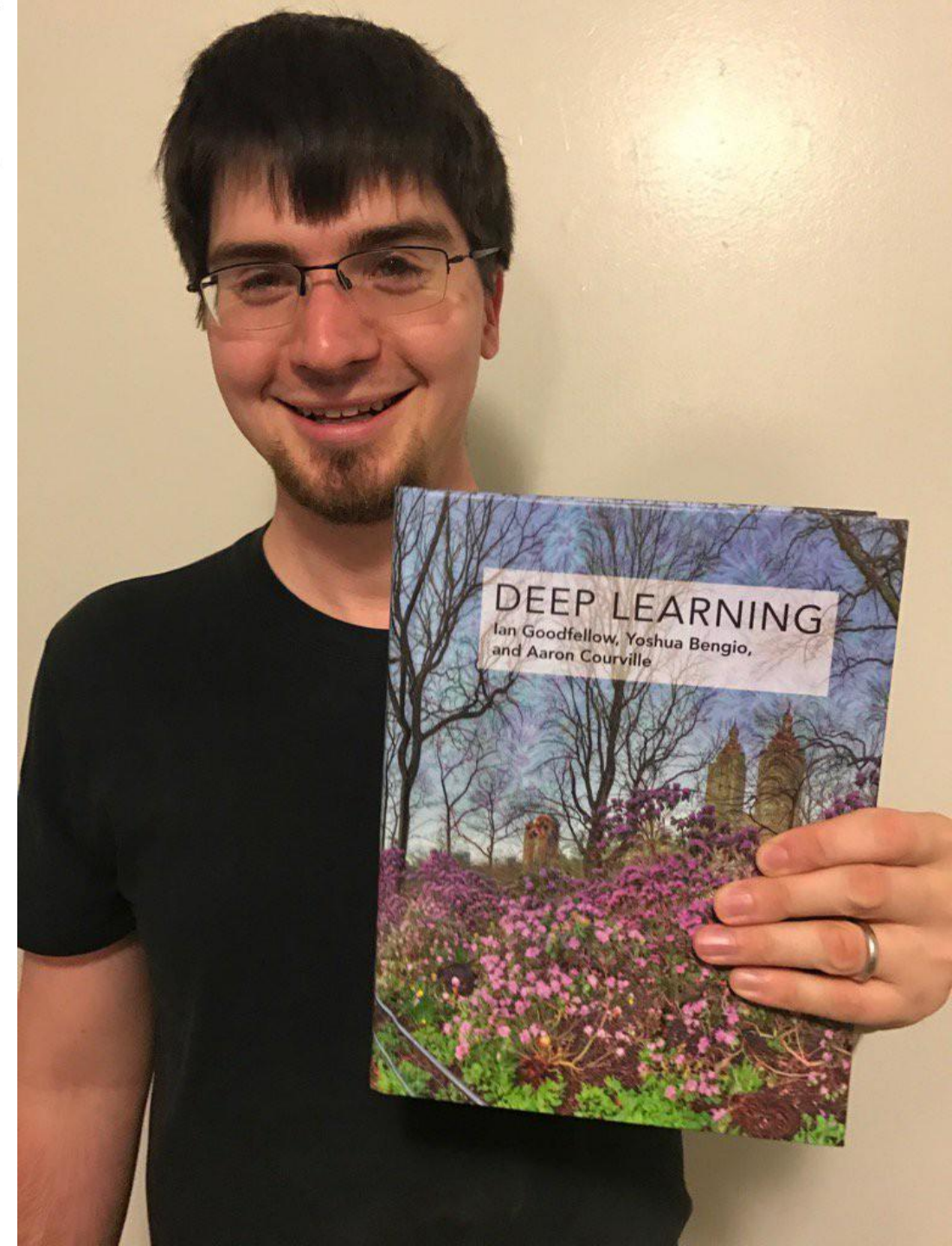
Generative Adversarial Nets

Ian J. Goodfellow, Jean Pouget-Abadie*, Mehdi Mirza, Bing Xu, David Warde-Farley,
Sherjil Ozair[†], Aaron Courville, Yoshua Bengio[‡]

Département d'informatique et de recherche opérationnelle
Université de Montréal
Montréal, QC H3C 3J7

Abstract

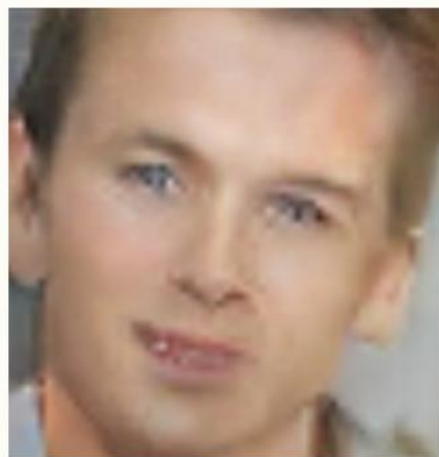
We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G . The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D , a unique solution exists, with G recovering the training data distribution and D equal to $\frac{1}{2}$ everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.



3.5 Years of Progress on Faces



2014



2015



2016



2017

(Brundage et al, 2018)

Progress Synthesizing Images

2014



GANs [Goodfellow et al.]

2015



DCGAN [Radford et al.]

2017



PG-GAN [Karras et al.]

2018

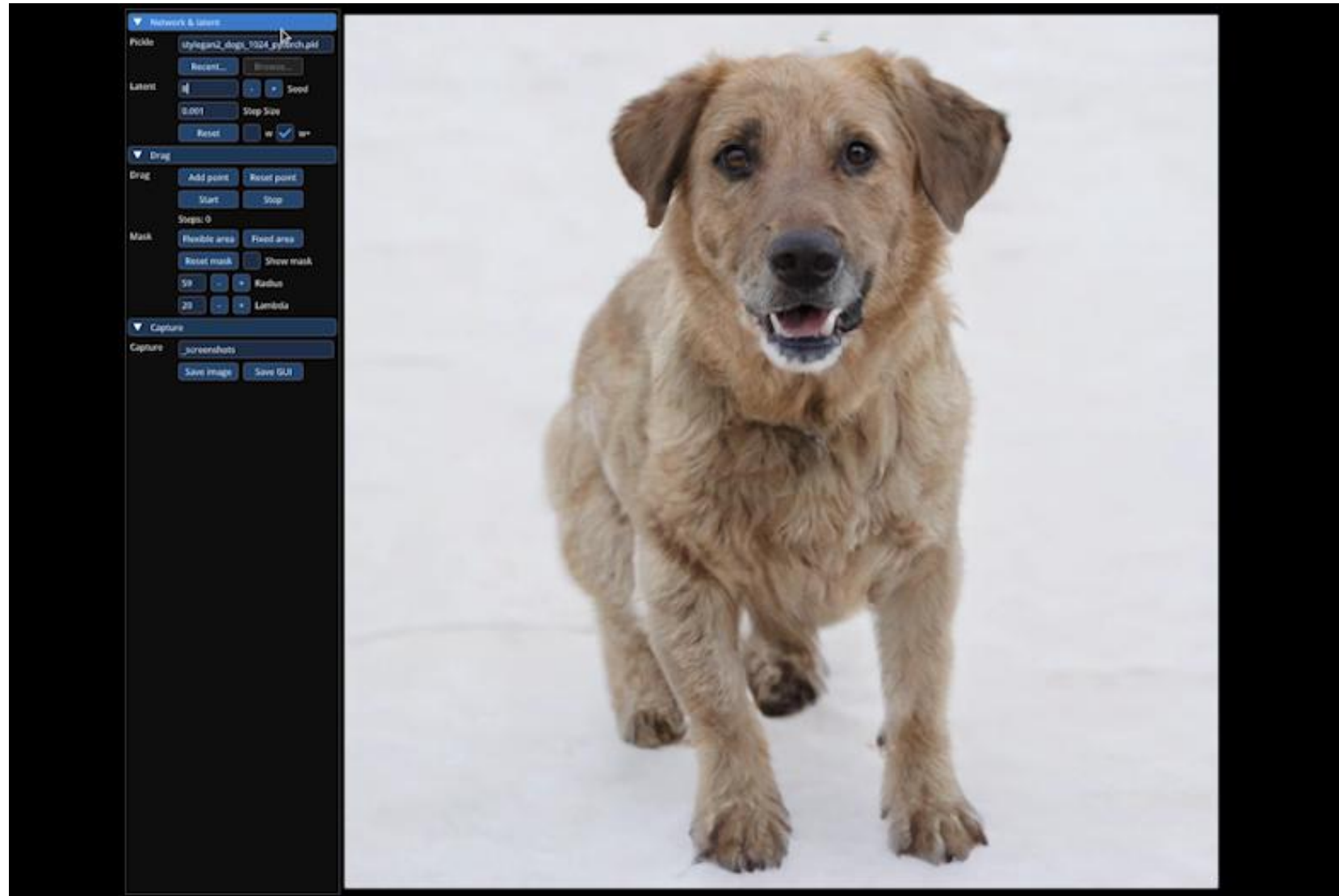


BigGAN [Brock et al.]

And Now...

Drag GAN

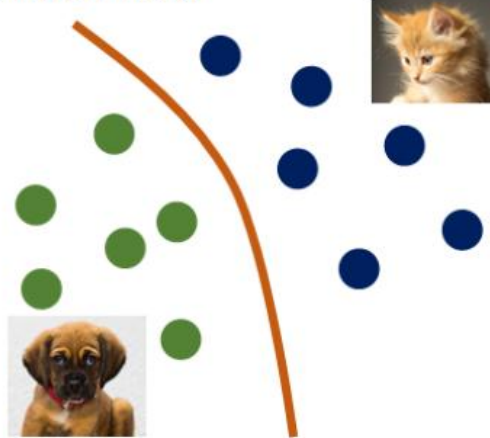
(2023)



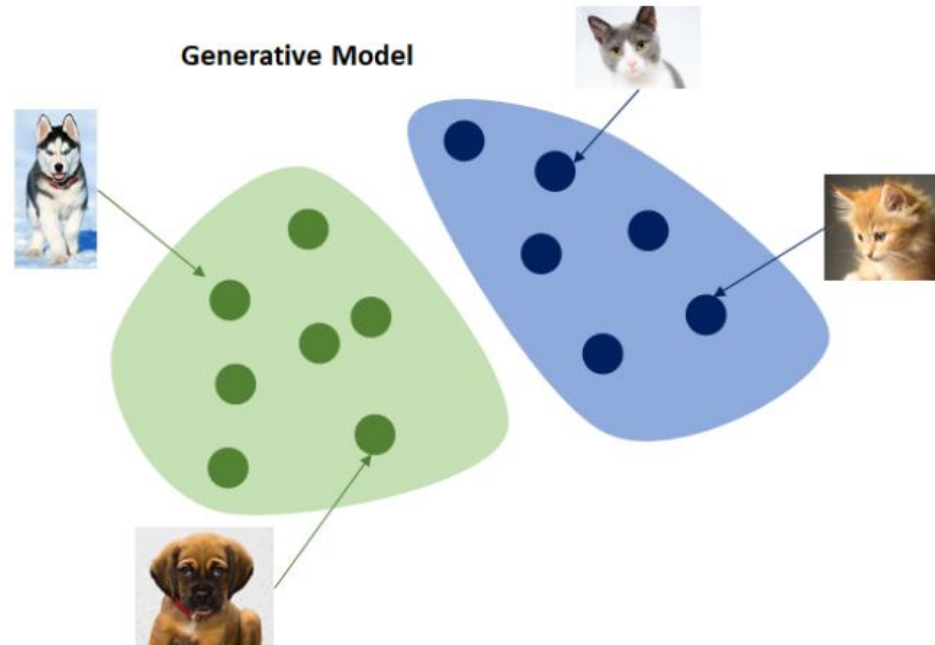
Discriminative vs Generative Models

- Machine learning models can be classified into two types: Discriminative and Generative.
- A **discriminative model** makes predictions on the unseen data based on conditional probability and can be used for classification or regression problems.
- A **generative model** focuses on the latent distribution of a dataset to return a probability for an example.

Discriminant Model



Generative Model



Generative Modeling: Sample Generation



Training Data
(CelebA)



Sample Generator
(Karras et al, 2017)

FPS 092.01

STAGE 1 00'16"18

40

PUSH P2 START

LAW

PAUL

2-Player Game (Minimax Game)

VS

Generator

Discriminator

2-Player Game

(Minimax Game)

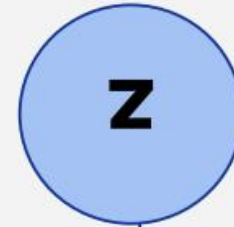
Discriminator

Learns to distinguish between real and generated data.

VS

Generator

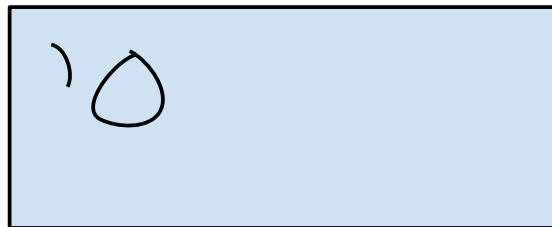
Learns to generate data to “fool” the discriminator.



VS



Generated Data



Discriminator

FAKE

REAL

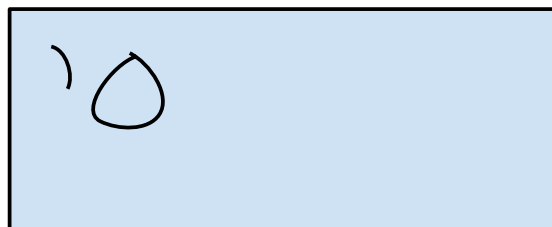
Real Data



Generated Data

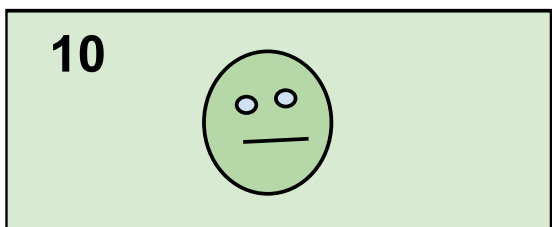
Discriminator

Real Data



FAKE

REAL



FAKE

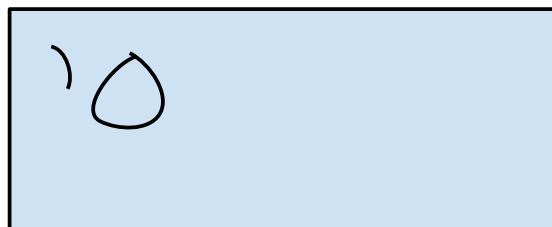
REAL



Generated Data

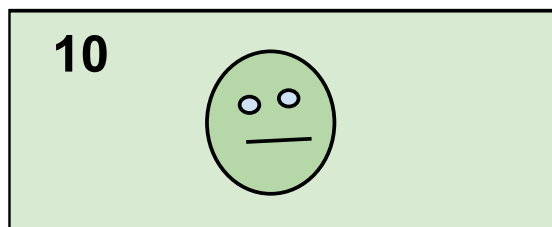
Discriminator

Real Data



FAKE

REAL



FAKE

REAL

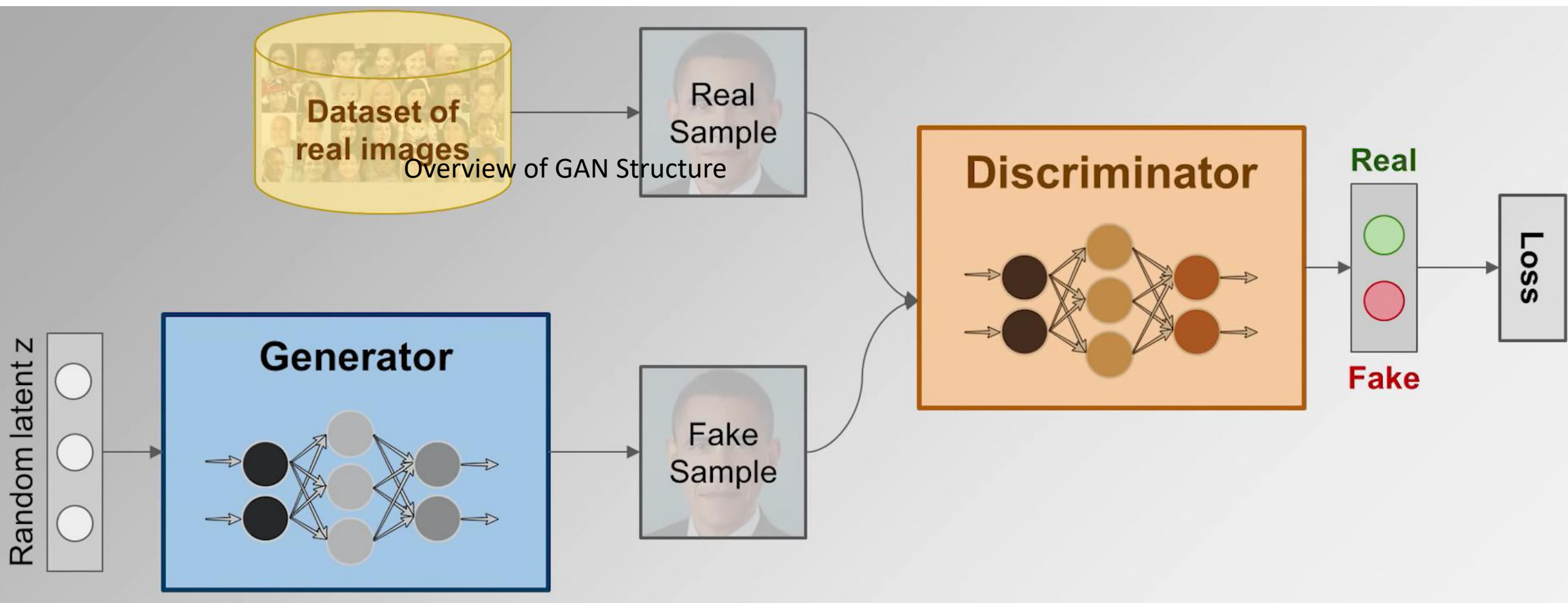


REAL

REAL



GANs Structure



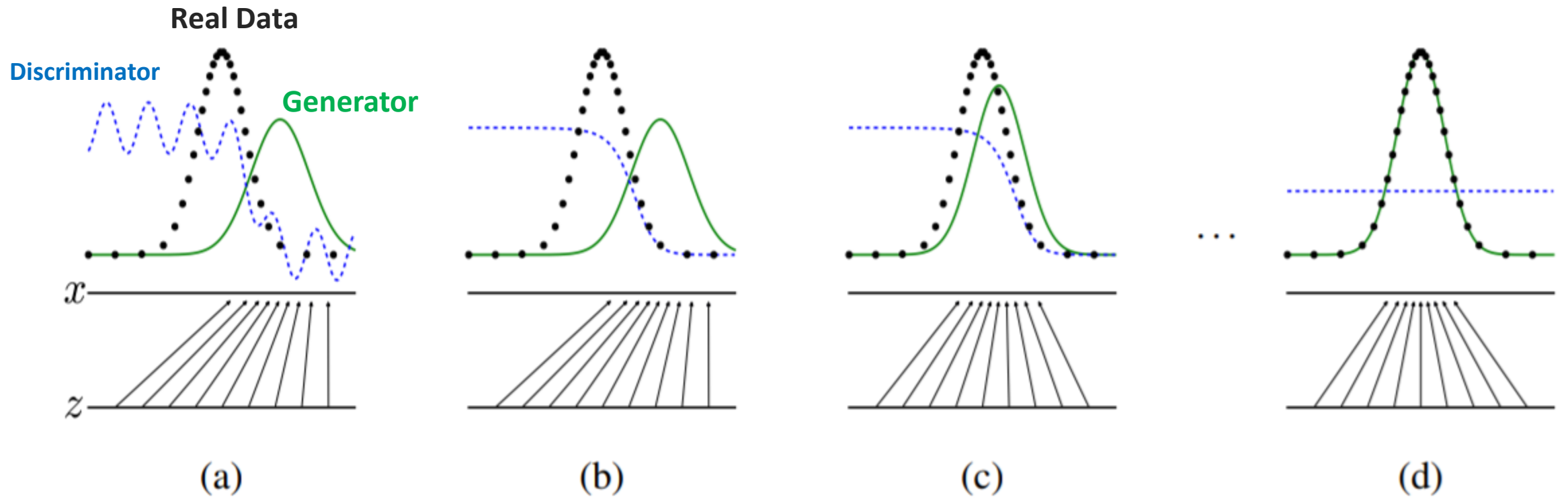
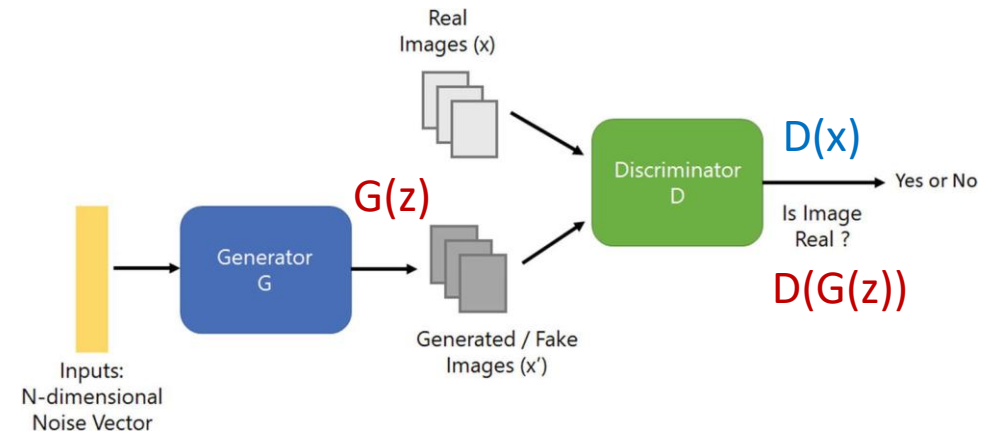


Figure 1: Generative adversarial nets are trained by simultaneously updating the discriminative distribution (D , blue, dashed line) so that it discriminates between samples from the data generating distribution (black, dotted line) p_x from those of the generative distribution p_g (G) (green, solid line). The lower horizontal line is the domain from which z is sampled, in this case uniformly. The horizontal line above is part of the domain of x . The upward arrows show how the mapping $x = G(z)$ imposes the non-uniform distribution p_g on transformed samples. G contracts in regions of high density and expands in regions of low density of p_g . (a) Consider an adversarial pair near convergence: p_g is similar to p_{data} and D is a partially accurate classifier. (b) In the inner loop of the algorithm D is trained to discriminate samples from data, converging to $D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)}$. (c) After an update to G , gradient of D has guided $G(z)$ to flow to regions that are more likely to be classified as data. (d) After several steps of training, if G and D have enough capacity, they will reach a point at which both cannot improve because $p_g = p_{\text{data}}$. The discriminator is unable to differentiate between the two distributions, i.e. $D(x) = \frac{1}{2}$.

Loss Function



$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))]$$

log-probability that D correctly predicts real data $\mathbf{x}$ are real

log-probability that D correctly predicts generated data $G(\mathbf{z})$ are generated

discriminator's (D) goal: **maximize** prediction accuracy

generator's (G) goal: **minimize** D's prediction accuracy, by **fooling** D into believing its outputs $G(\mathbf{z})$ are real as often as possible

Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k , is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations **do**

for k steps **do**

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Sample minibatch of m examples $\{x^{(1)}, \dots, x^{(m)}\}$ from data generating distribution $p_{\text{data}}(x)$.
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D(x^{(i)}) + \log \left(1 - D(G(z^{(i)})) \right) \right].$$

end for

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log \left(1 - D(G(z^{(i)})) \right).$$

end for

The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

GANs in Nature

...

۲۰۱۸ نوامبر ۱۱ • @Reza_Zadeh  **Reza Zadeh**



GANs in nature: Cuckoos (Generator) lay eggs in Warbler (Discriminator) bird's nests, tricking warblers to raise cuckoo chicks. Warblers are evolving to recognize cuckoo eggs to destroy them. Simultaneously, cuckoos evolve to produce more warbler-like eggs cam.ac.uk/research/featu...



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Deep Convolutional GANs (DCGAN)

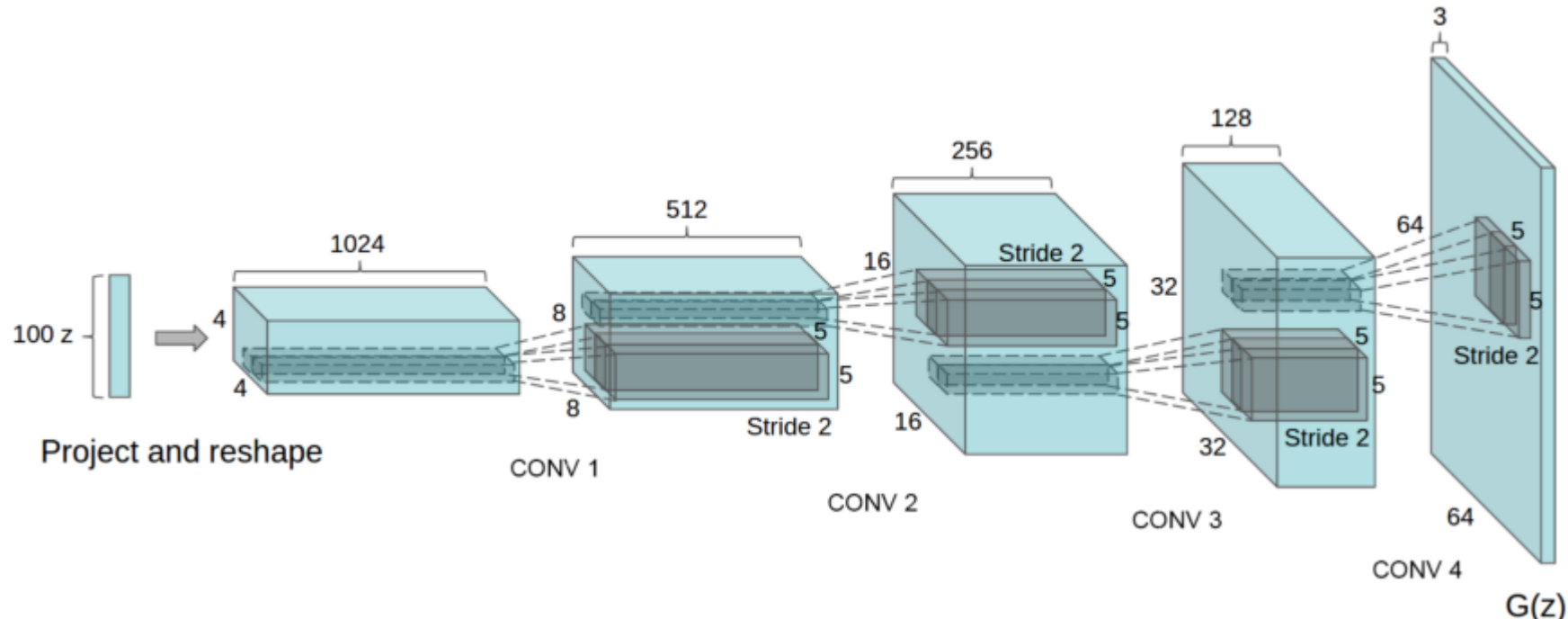
Want to learn more?



Radford, et al. **Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks**. International Conference on Learning Representations (2016)

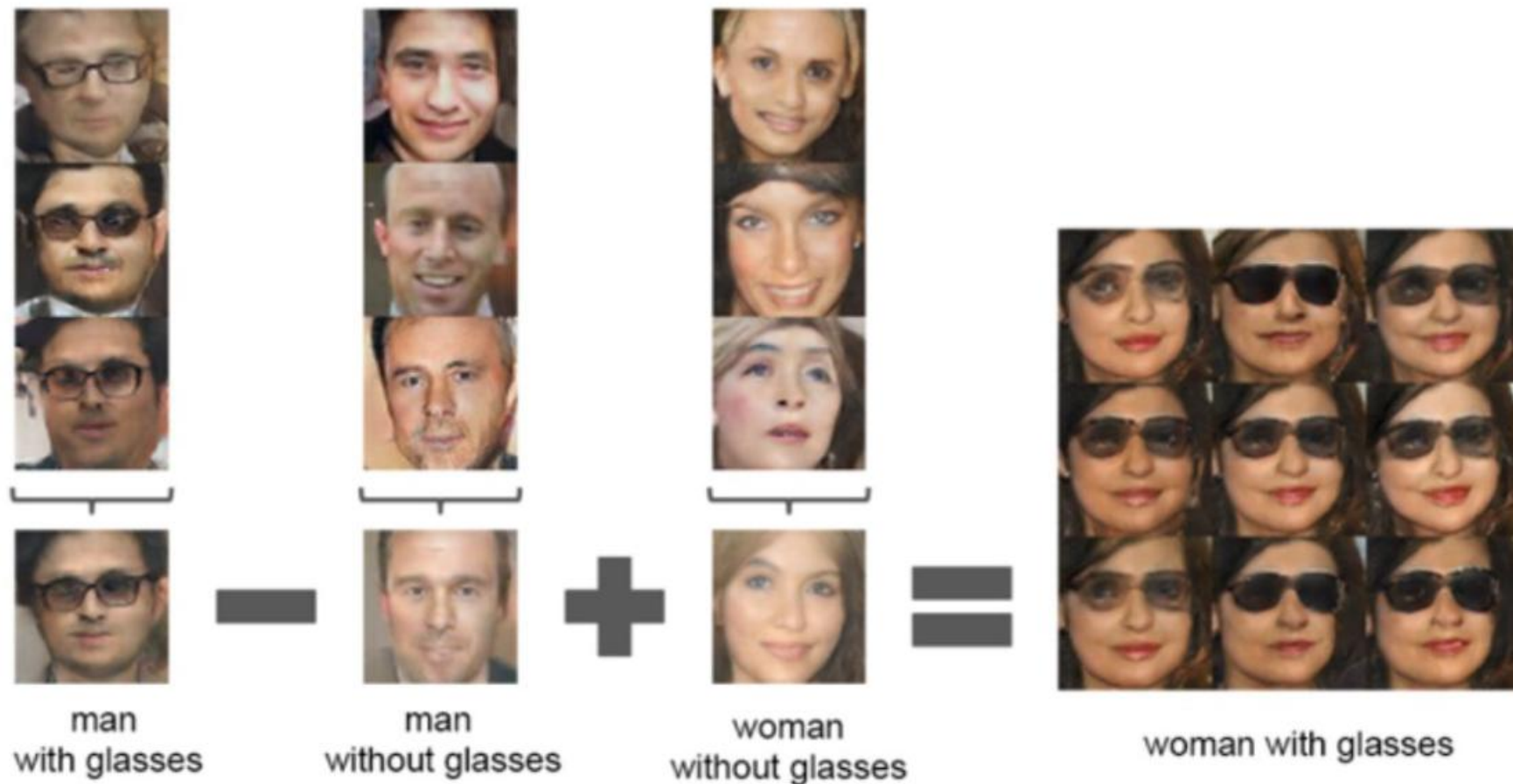
Simply use deep convnets for G and D

- Importantly, **batch normalization** (Ioffe and Szegedy, 2015) helped to stabilize the difficult learning process



Deep Convolutional GANs (DCGAN)

The DCGAN generator's noise/latent space appears to have **meaningful semantics**



Conditioning information for training GANs

Want to learn more?



Mirza and Osindero. **Conditional Generative Adversarial Nets**.
arXiv:1411.1784 (2014)

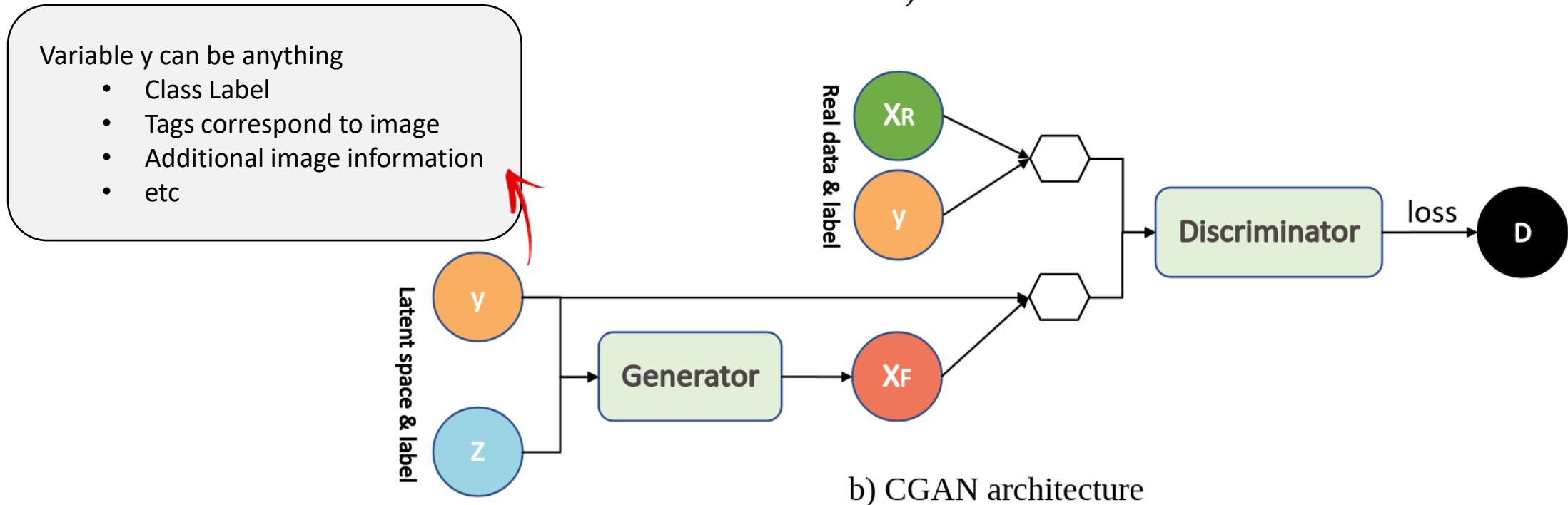
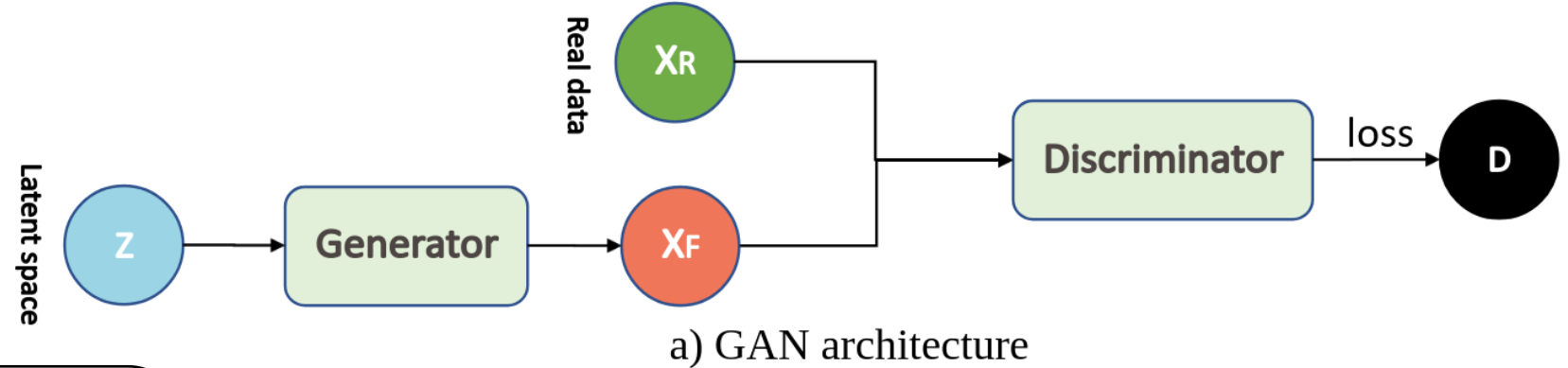
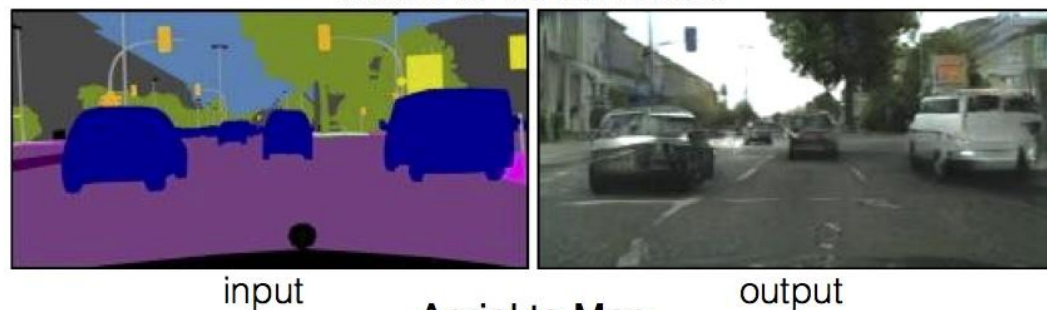
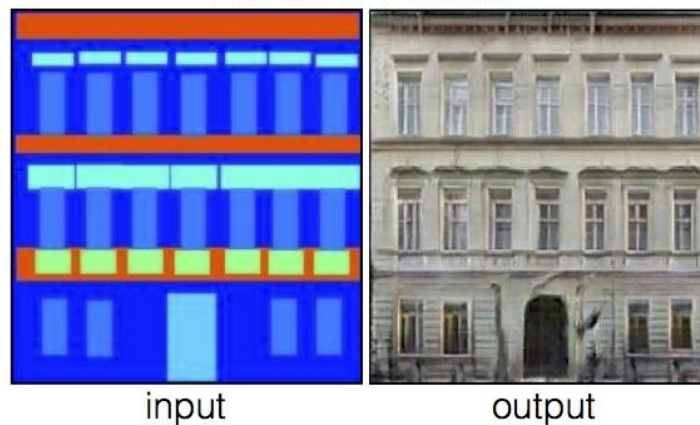


Image-to-Image Translation with Conditional Adversarial Nets

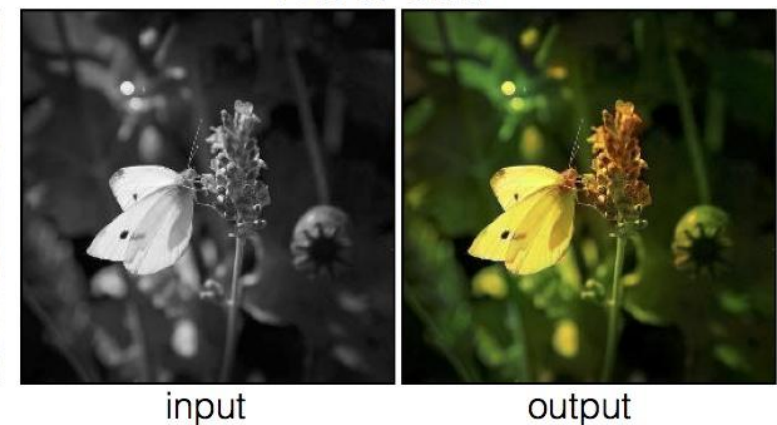
Labels to Street Scene



Labels to Facade



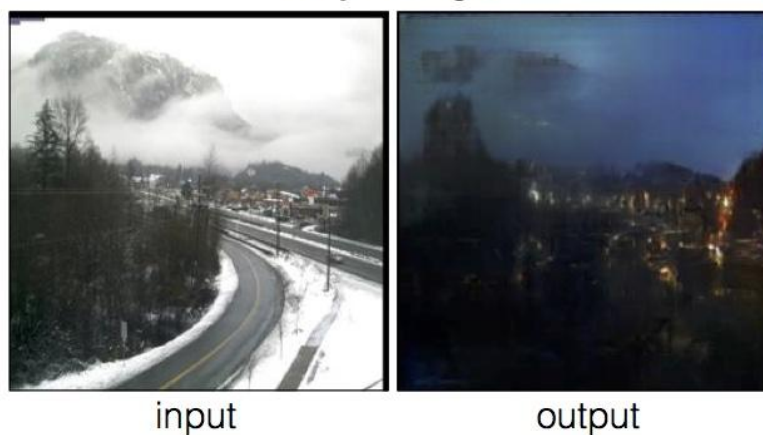
BW to Color



Aerial to Map



Day to Night



Edges to Photo



Text to Image Synthesis

This bird has a yellow belly and tarsus, grey back, wings, and brown throat, nape with a black face



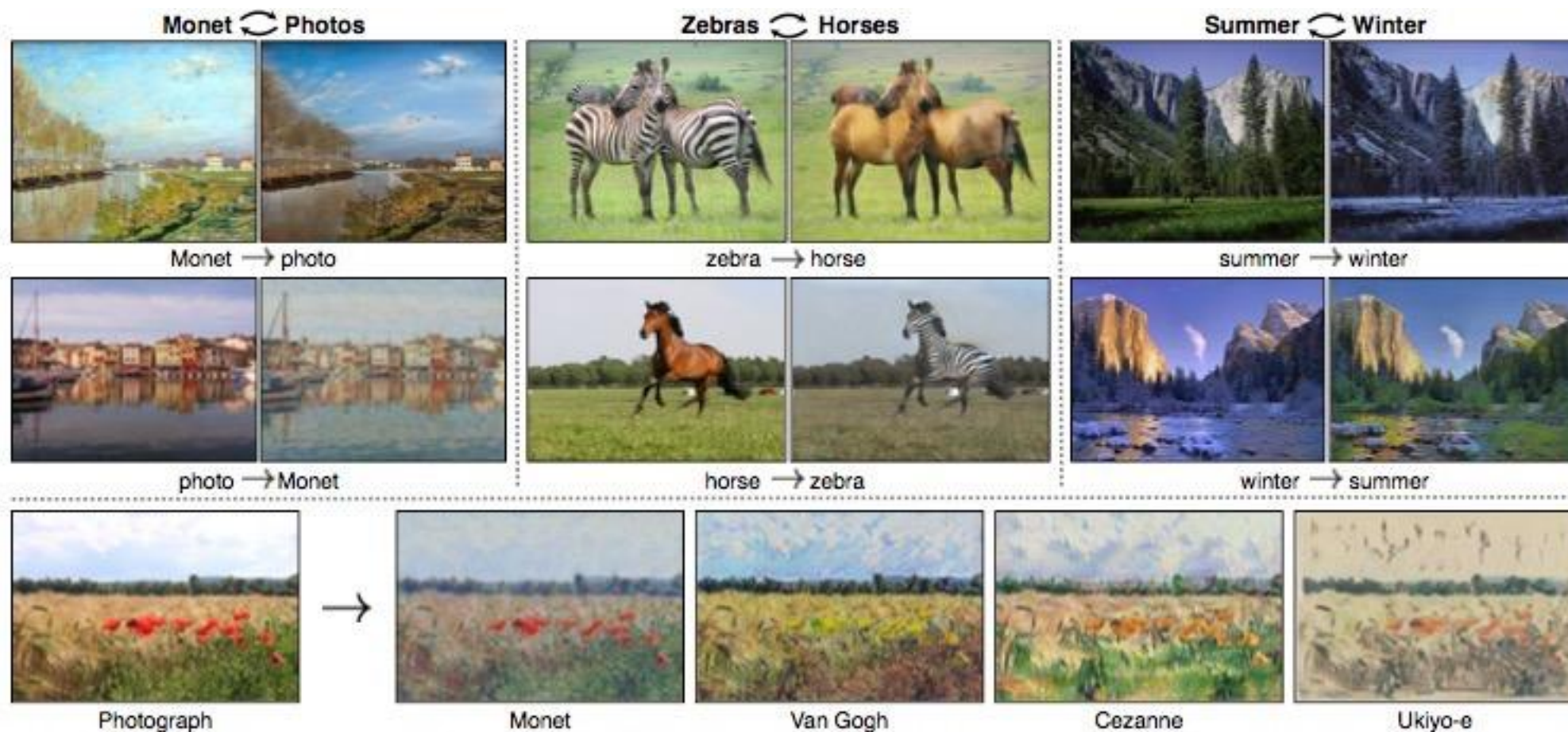
(Zhang et al., 2016)

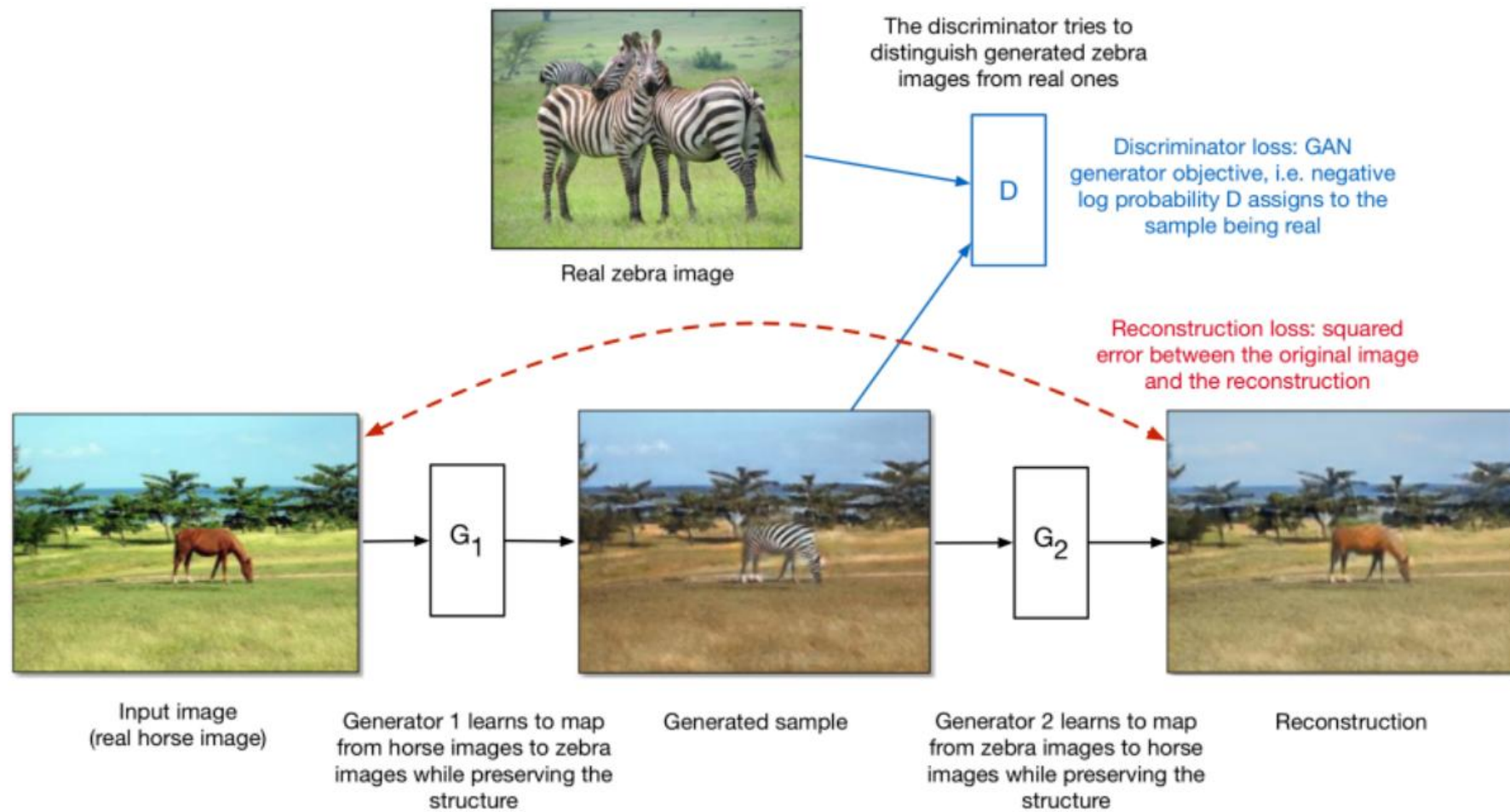
Cycle GANs

Want to learn more?



Zhu, et al. Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. International Conference on Computer Vision (2017)

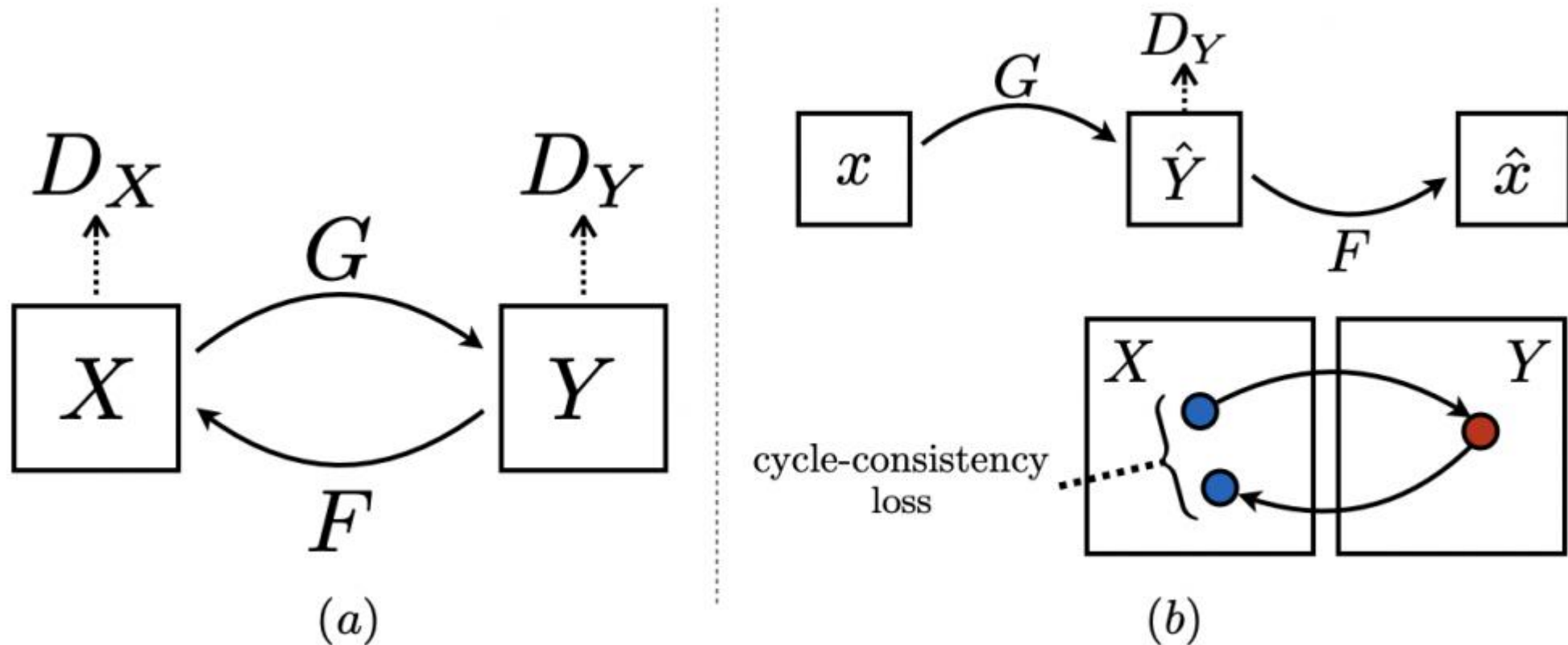




$$\text{Total loss} = \text{discriminator loss} + \text{reconstruction loss}$$

Cycle GANs

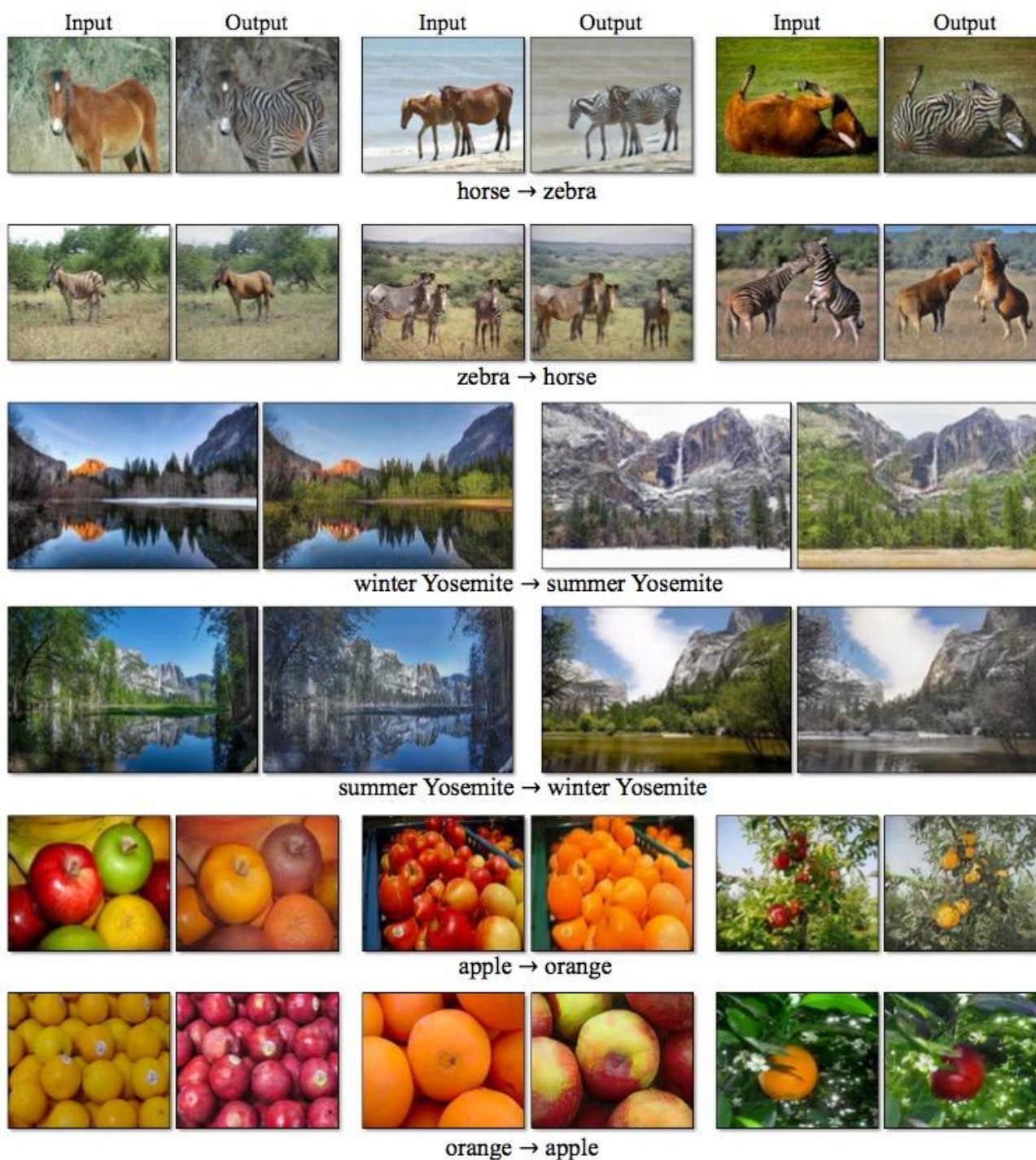
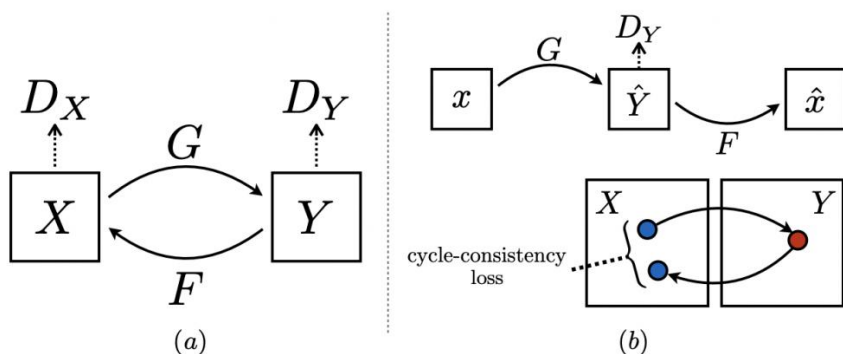
- Train a generator to **translate** between images of two different domains
But **without any paired samples!**



Cycle GANs

Enforces **cycle consistency**:

- Image x in domain A
- Translate to domain B
- Back to domain A $\rightarrow x'$
- Enforce $x \approx x'$



Photos to paintings



Paintings to photos



Super Resolution GANs (SRGAN)



Figure 2: From left to right: bicubic interpolation, deep residual network optimized for MSE, deep residual generative adversarial network optimized for a loss more sensitive to human perception, original HR image. Corresponding PSNR and SSIM are shown in brackets. [4× upscaling]

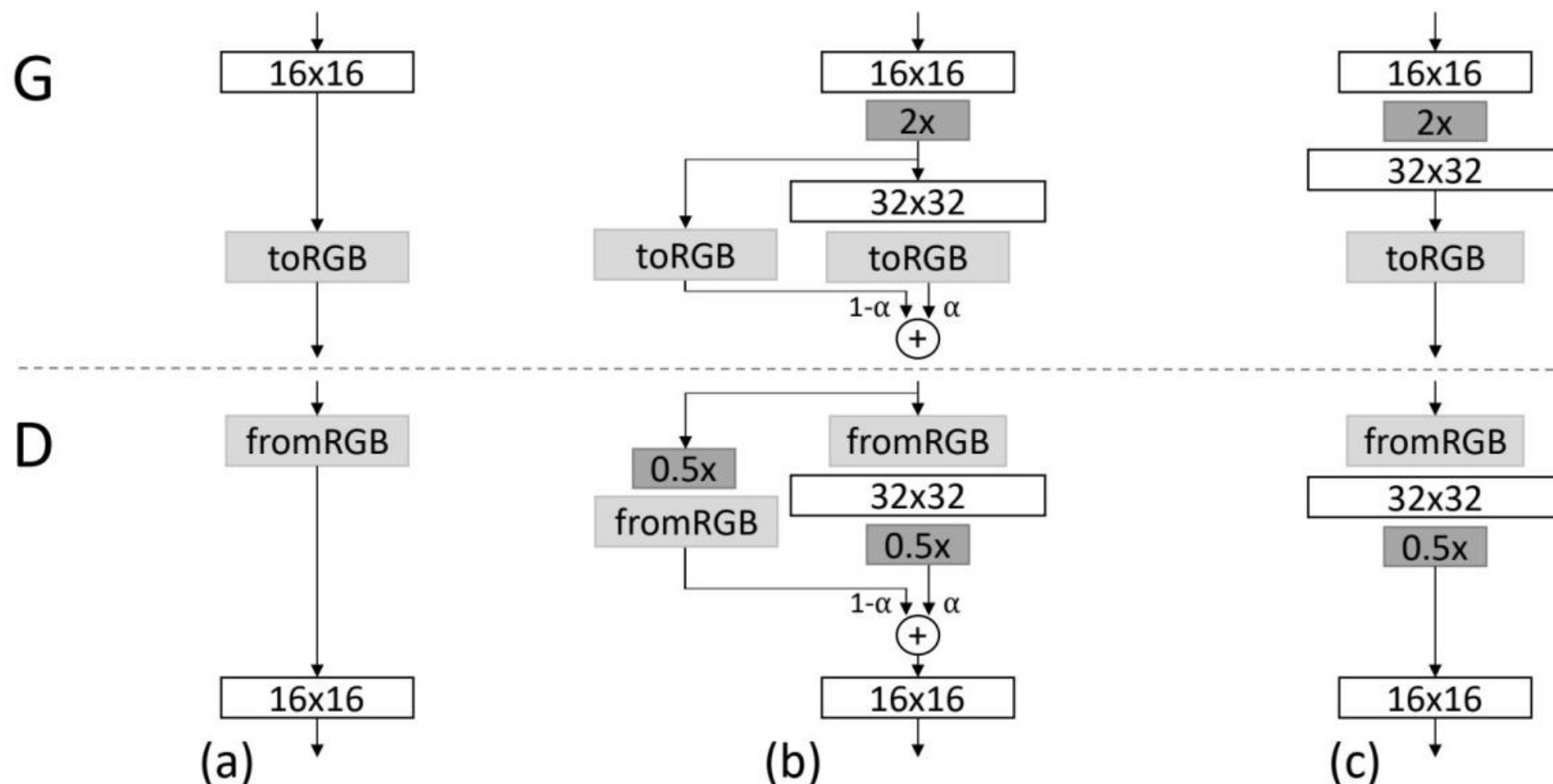
Progressive GANs

Want to learn more?



Karras, et al. **Progressive Growing of GANs for Improved Quality, Stability, and Variation**. International Conference on Learning Representations (2018)

- First, train a GAN to generate tiny (4x4) images
- After convergence, add a new layer (in G & D) to generate 8x8 resolution images
- Repeat for 16x16, 32x32, ...
- Very compelling results in a restricted domain (faces)



Progressive GANs

Want to learn more?



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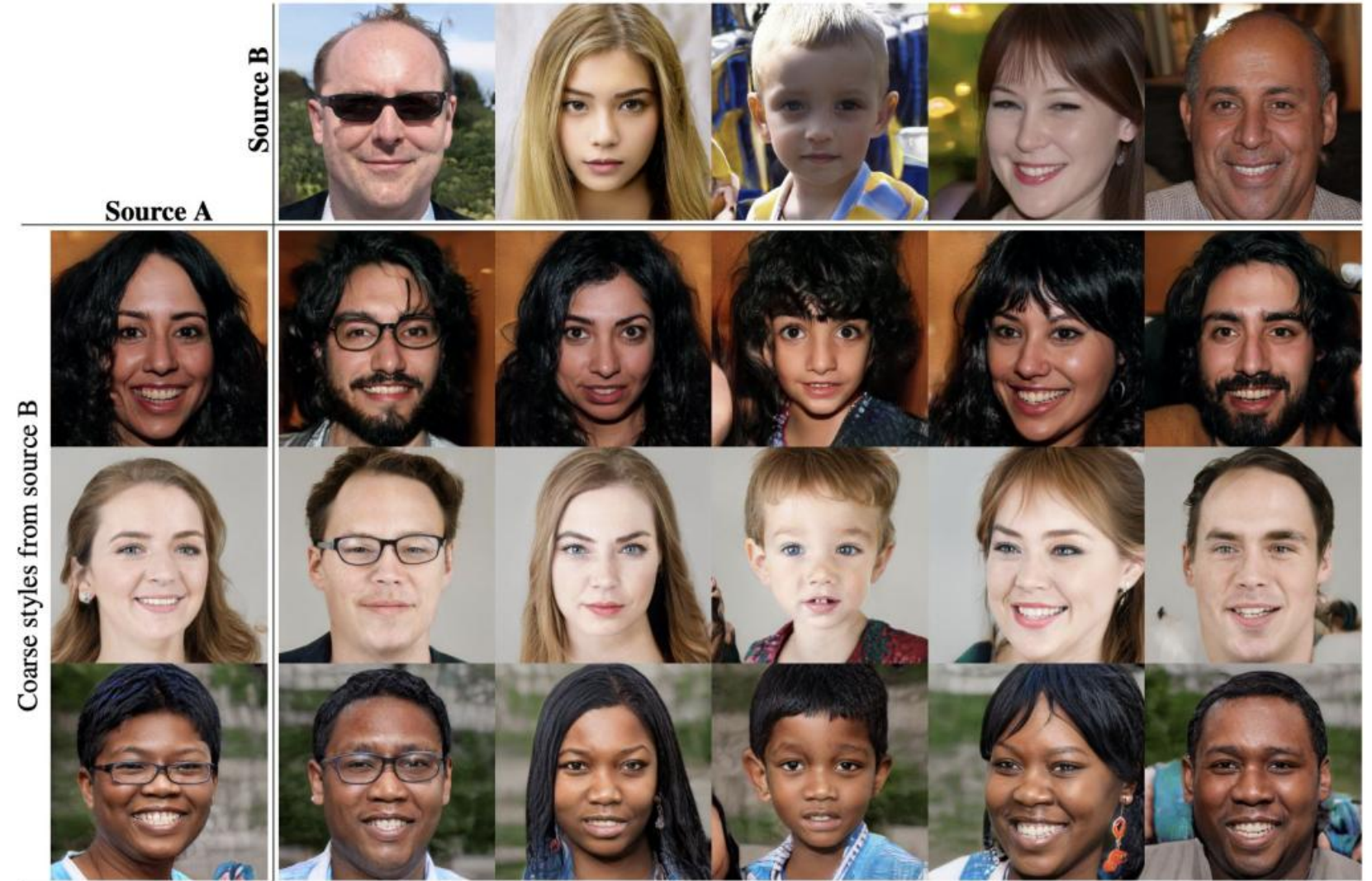
Style GANs

Want to learn more?



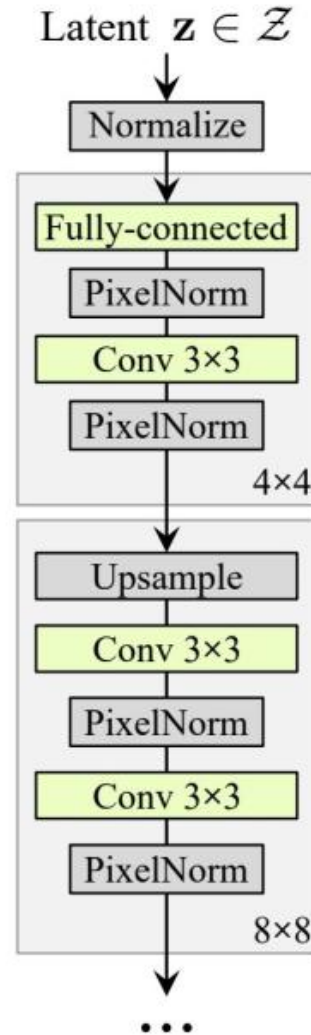
Karras, et al. A Style-Based Generator Architecture for Generative Adversarial Networks. IEEE Conference on Computer Vision and Pattern Recognition (2019)

StyleGANs were shown to be capable of generating remarkably photo realistic face images
Structured latent inputs (z) to the generator can be used to control its outputs in various interesting ways.

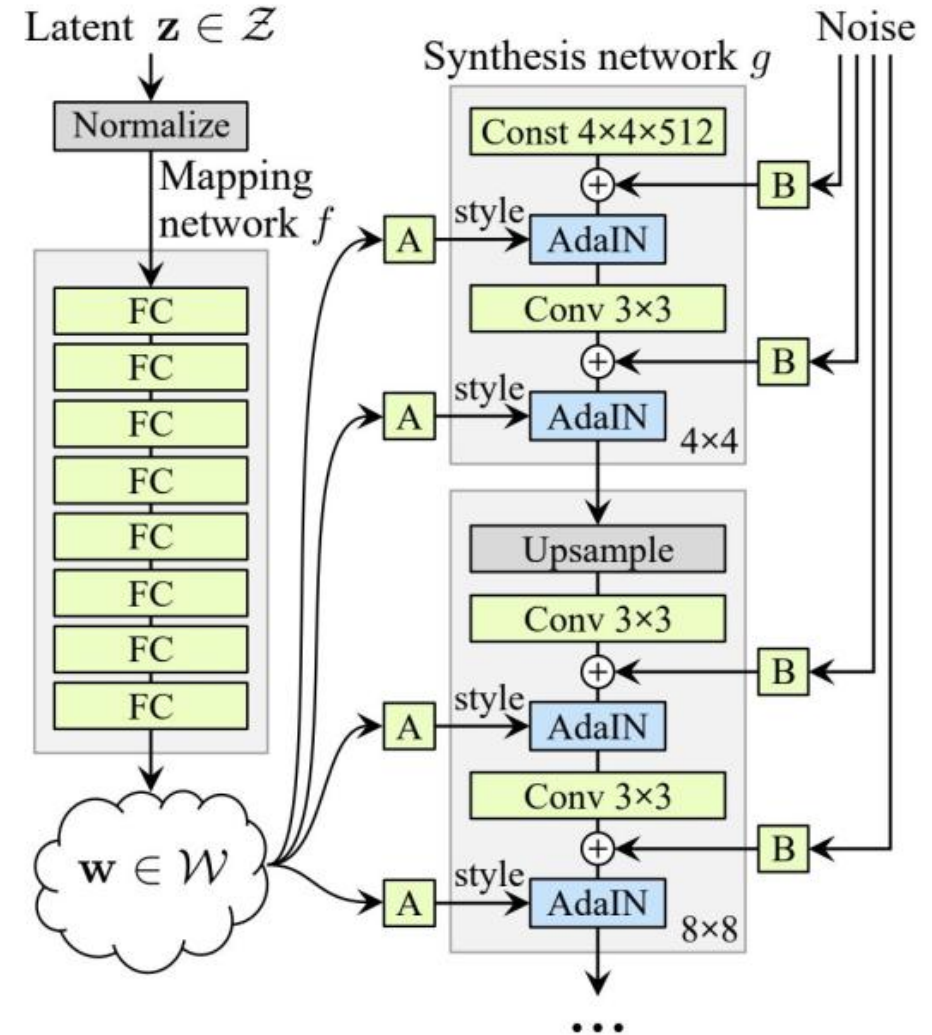


Style GANs

- Global latents transformed via an 8 layer MLP
- Incorporates spatial **pixel noise** at each layer
- Single-channel "image" of noise
- Broadcast via learnt per-channel scaling factors
- Model learns to associate global latents with the overall **style** of the image
- Pixel noise modulates the local appearance



(a) Traditional



(b) Style-based generator

Want to learn more?



Karras, et al. A Style-Based Generator Architecture for Generative Adversarial Networks. IEEE Conference on Computer Vision and Pattern Recognition (2019)

BigGANs

- Make GANs really big
 - Big batches
 - Big models
 - Big datasets
 - Big (high res) images
- Trained on ImageNet (1.2M images) and JFT (300M images)

Want to learn more?



Brock, et al. Large Scale GAN Training for High Fidelity Natural Image Synthesis. International Conference on Learning Representations (2019)



BigGANs

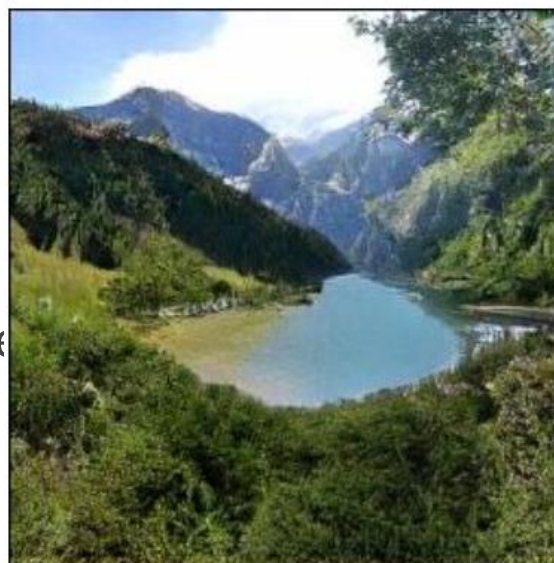
Large empirical study to build a reliable recipe for large scale GAN training, including:

- Hinge loss in D
- Spectral norm
- Self-attention
- Projection disc
- Orthogonal regularisation
- "Skip connections" from noise
- Class label embedding shared across layers

Want to learn more?



Brock, et al. Large Scale GAN Training for High Fidelity Natural Image Synthesis. International Conference on Learning Representations (2019)



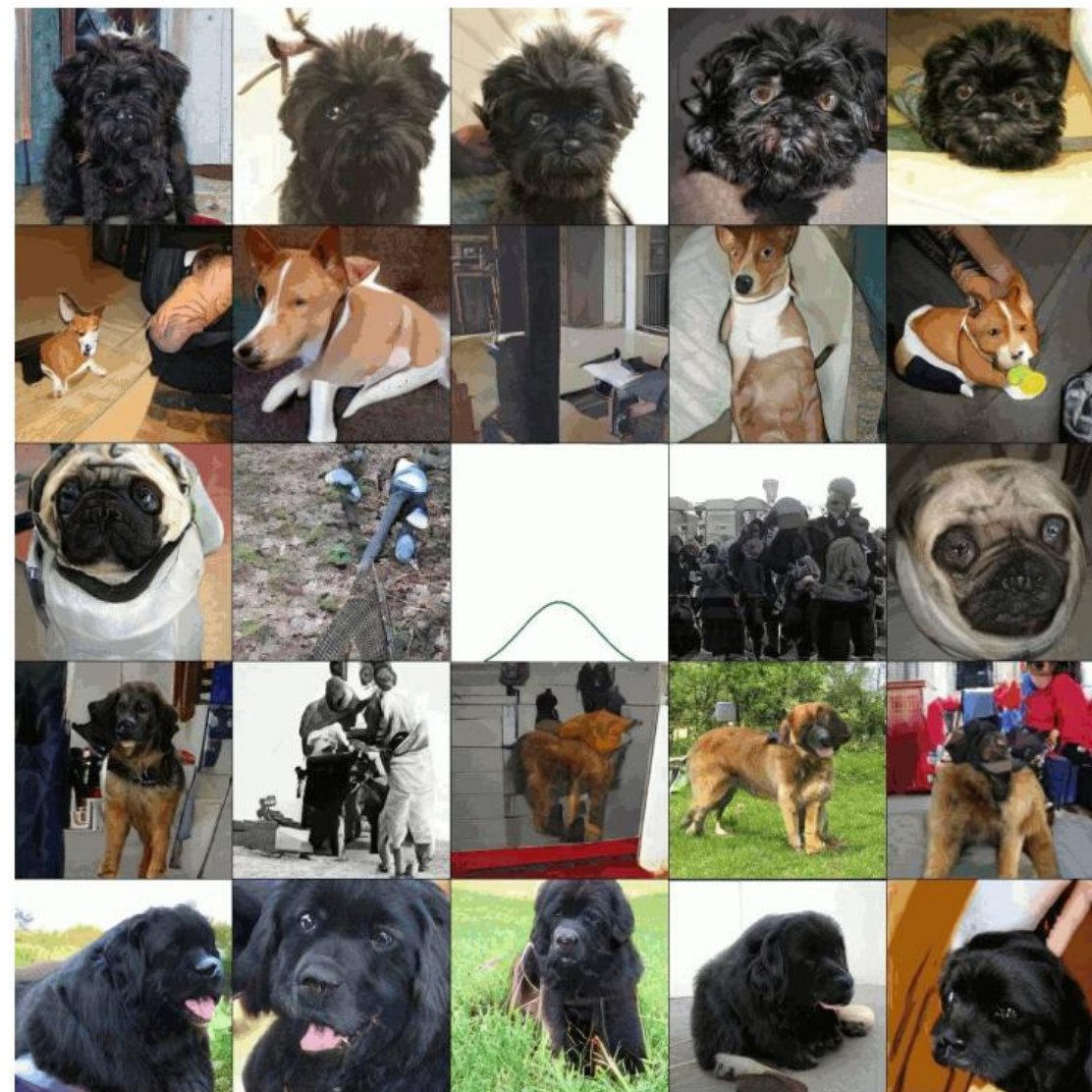
BigGANs

- Introduced the **truncation trick**
- Change the scale of the noise z input to the generator
- Make the noise smaller (truncate) to increase image fidelity
- Generates prototypical examples of each class
- Make the noise larger to increase variety
- Generates the full class distribution

Want to learn more?



Brock, et al. Large Scale GAN Training for High Fidelity Natural Image Synthesis. International Conference on Learning Representations (2019)



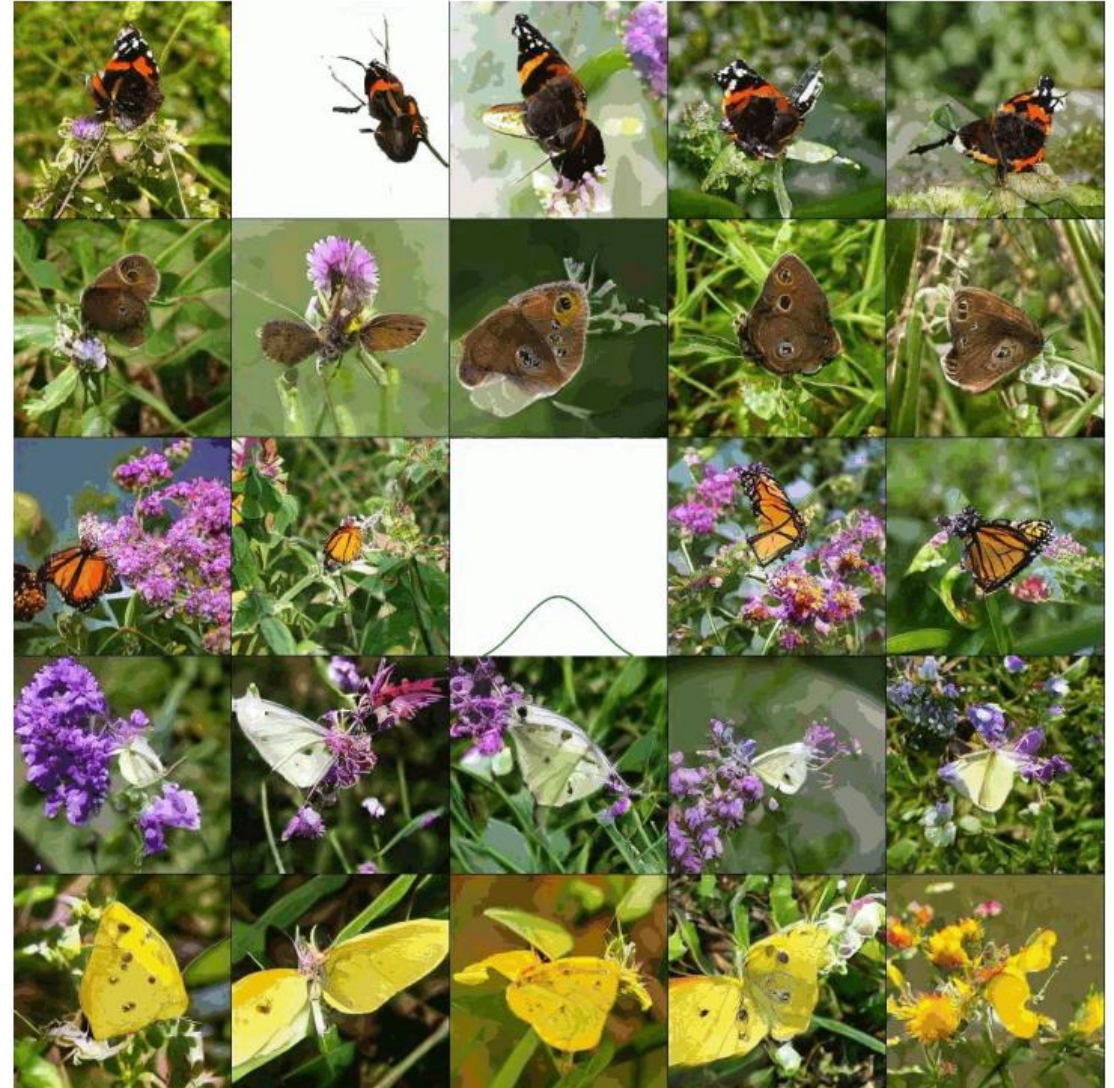
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Brock, et al. Large Scale GAN Training for High Fidelity Natural Image Synthesis. International Conference on Learning Representations (2019)

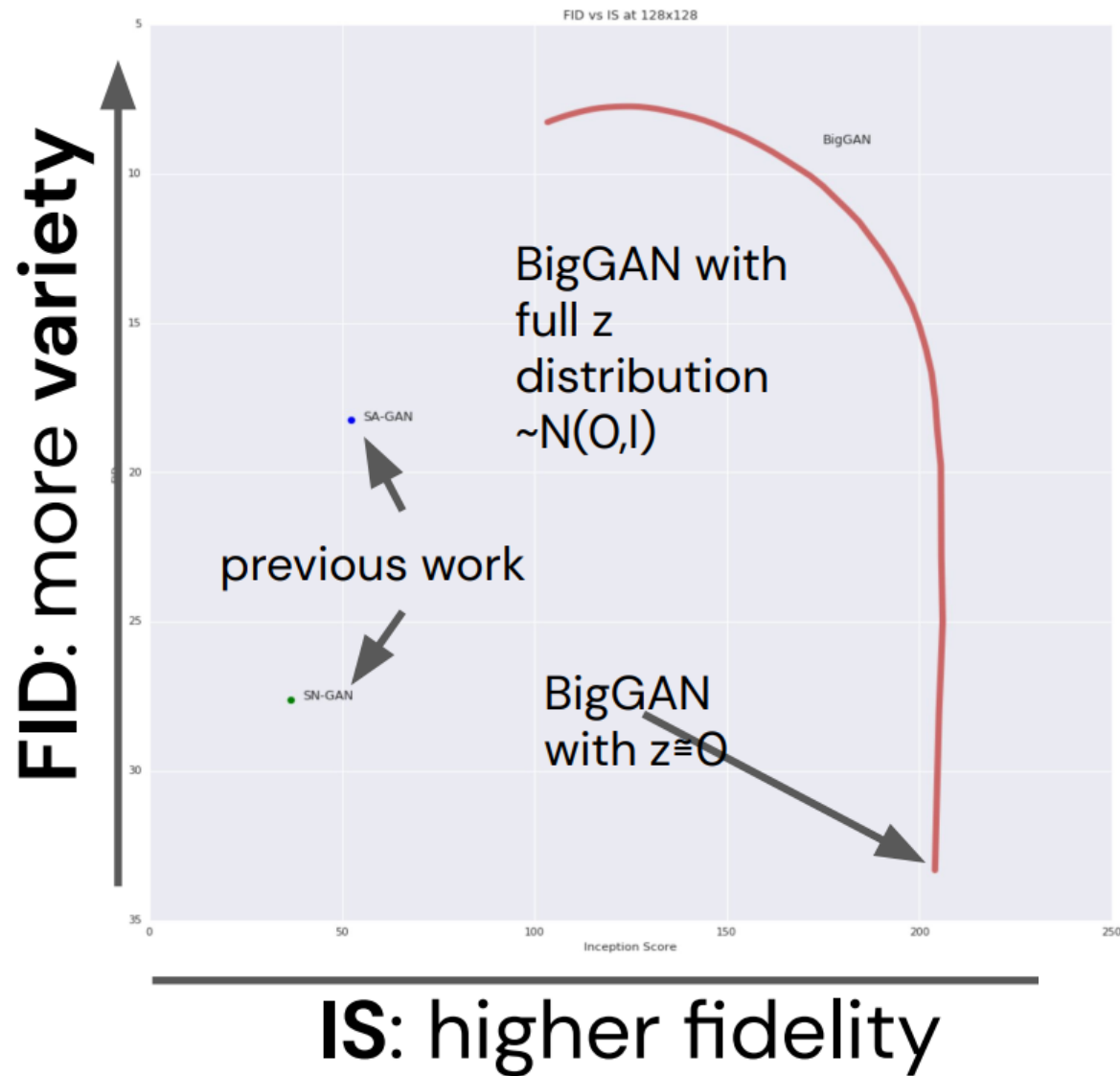
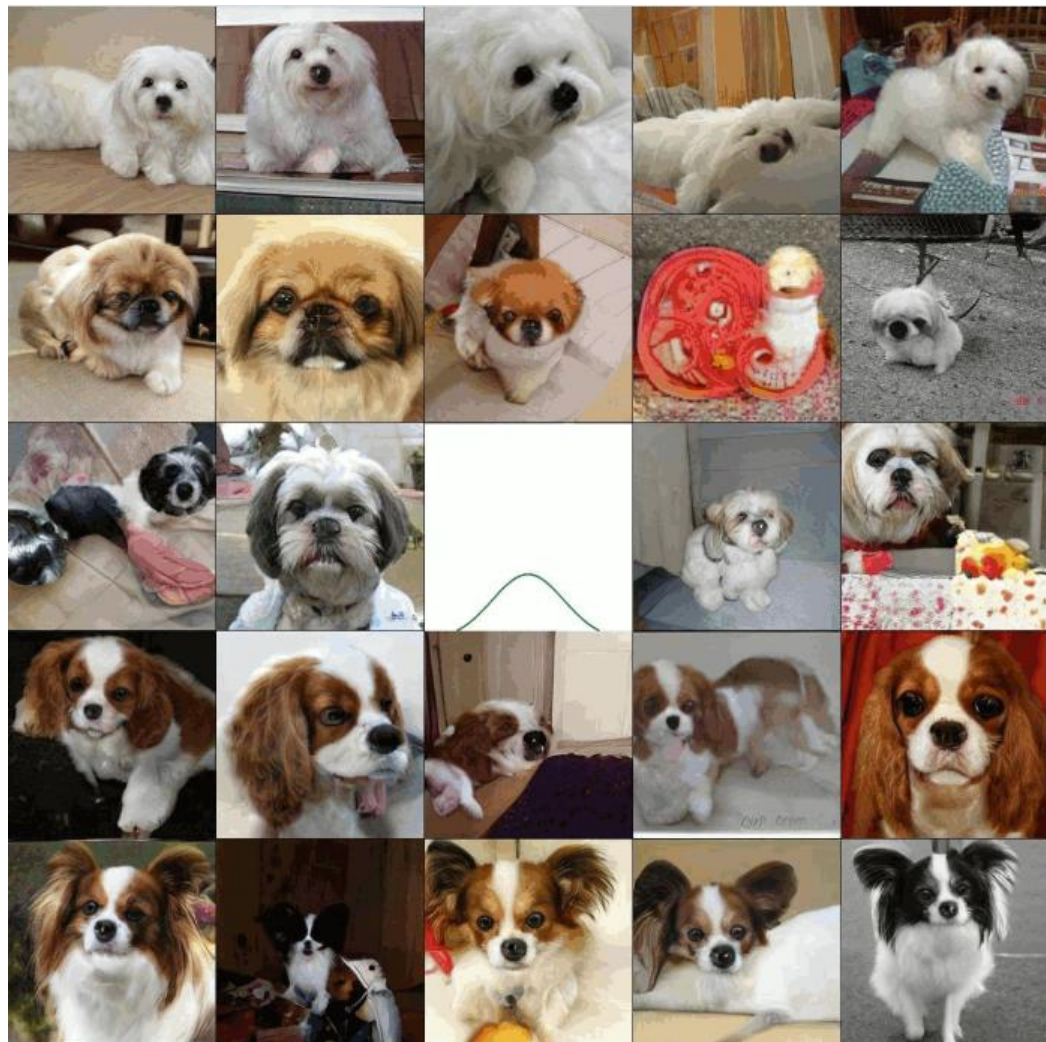


BigGANs

Want to learn more?



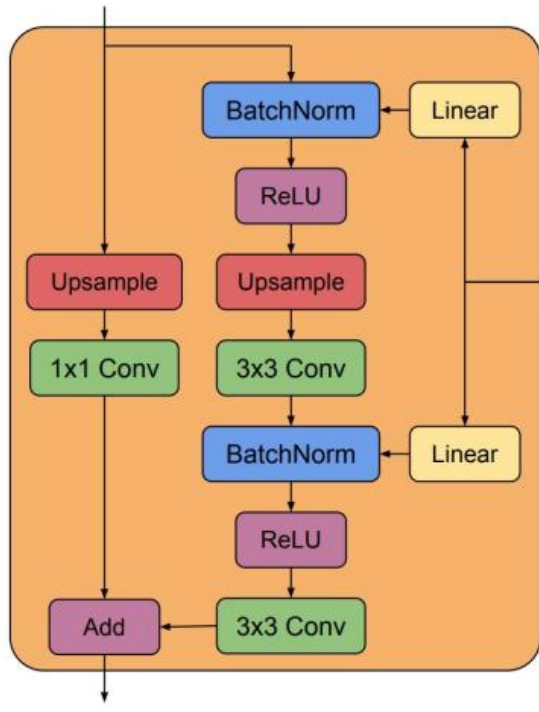
Brock, et al. Large Scale GAN Training for High Fidelity Natural Image Synthesis. International Conference on Learning Representations (2019)



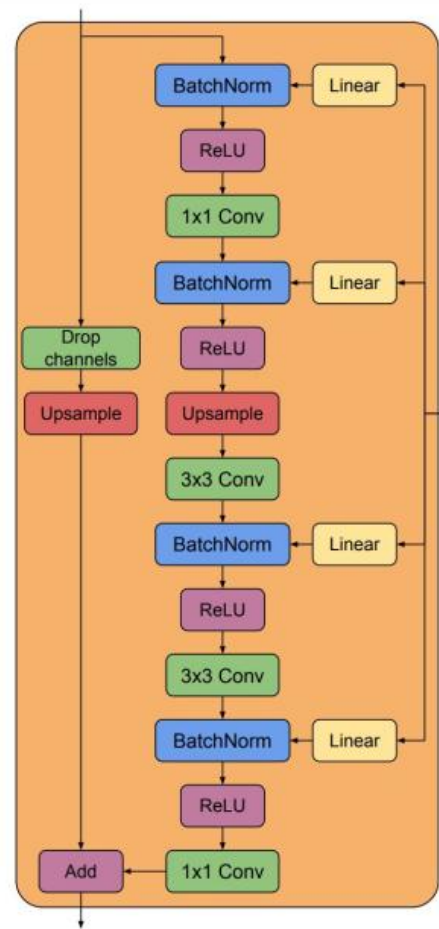
BigGANs

4x deeper, but more efficient!

BigGAN (original) Block



BigGAN-deep Block

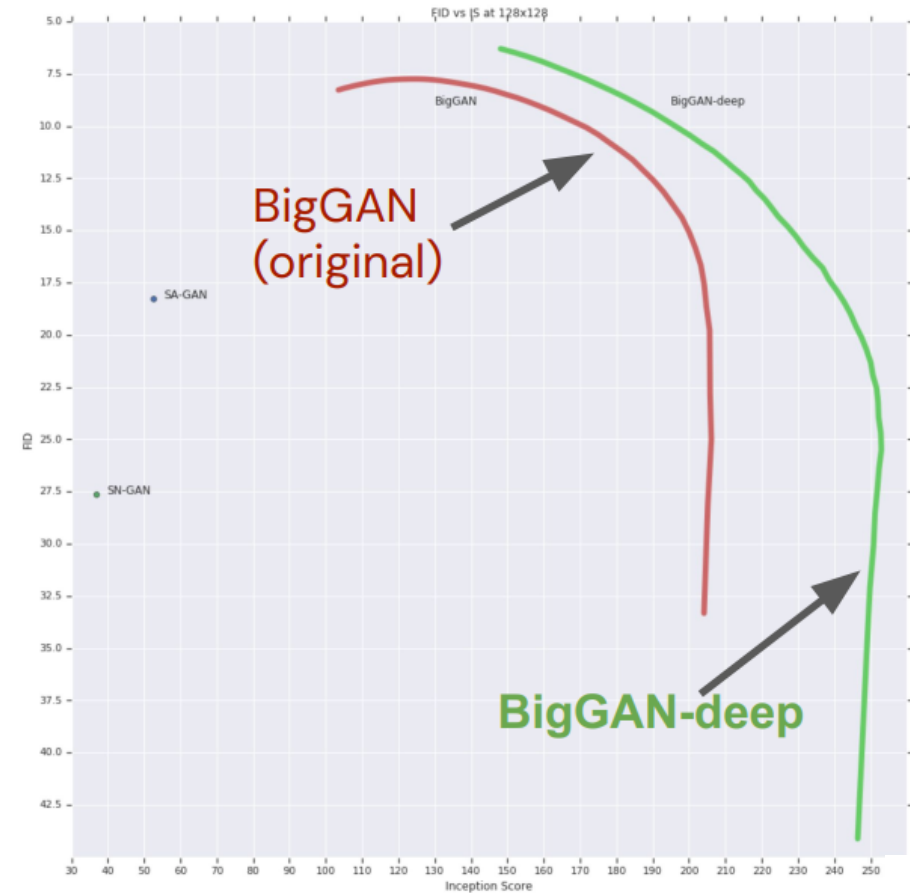


Want to learn more?



Brock, et al. Large Scale GAN Training for High Fidelity Natural Image Synthesis. International Conference on Learning Representations (2019)

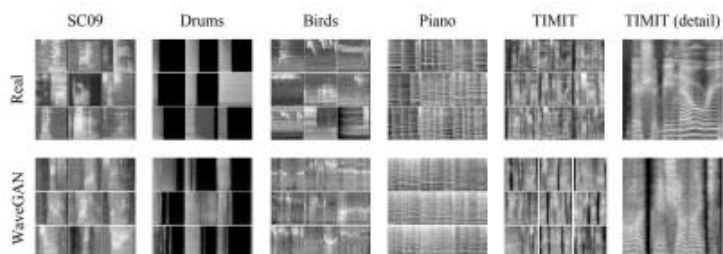
FID: more variety



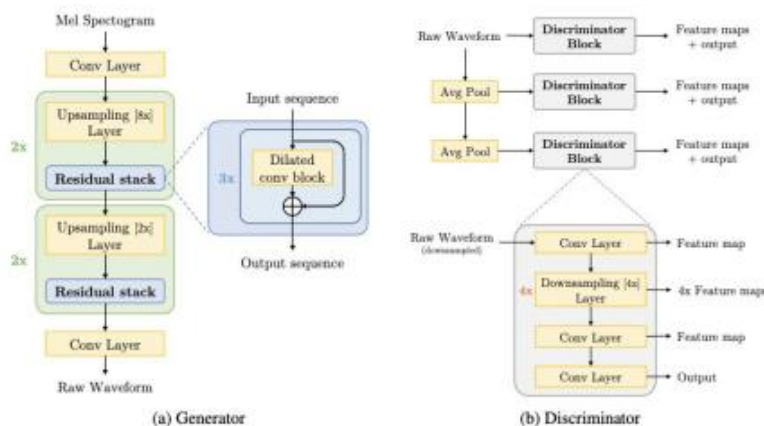
IS: higher fidelity

GANs for Audio Synthesis

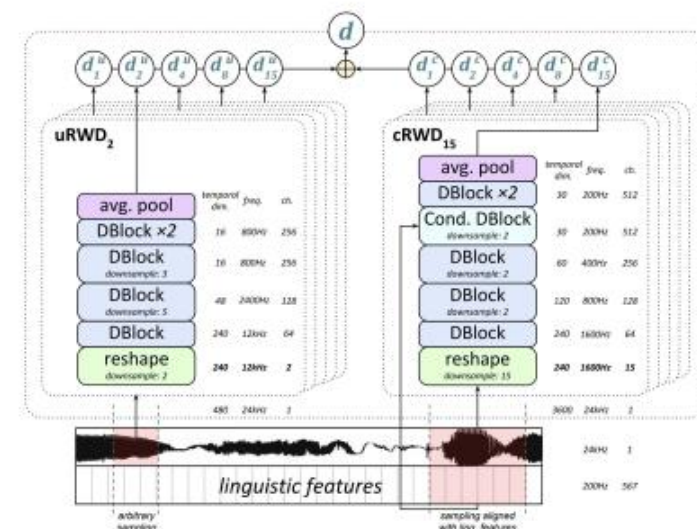
WaveGAN (C. Donahue et al.)



MelGAN (Kumar et al.)



GAN-TTS (Bińkowski et al.)



Want to learn more?



C. Donahue et al. **Adversarial Audio Synthesis**. International Conference on Learning Representations (2019)

Want to learn more?



Kumar et al. **MelGAN: Generative Adversarial Networks for Conditional Waveform Synthesis**. Neural Information Processing Systems (2019)

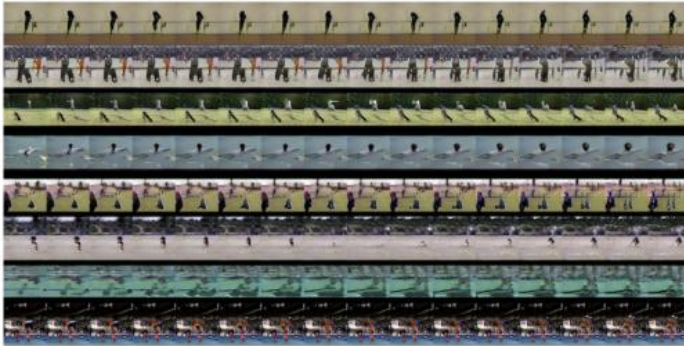
Want to learn more?



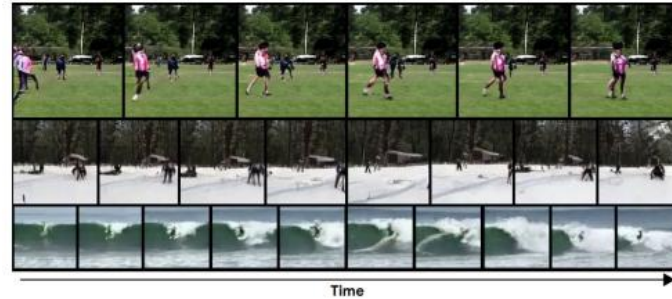
Bińkowski et al. **High Fidelity Speech Synthesis with Adversarial Networks**. International Conference on Learning Representations (2020)

GANs for Video Synthesis & Prediction

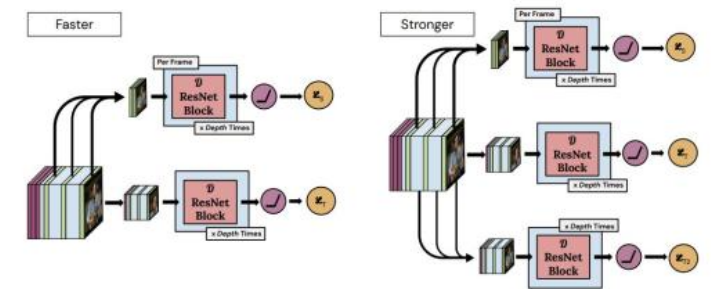
TGAN-v2 (Saito & Saito)



DVD-GAN (Clark et al.)



TriVD-GAN (Luc et al.)



Want to learn more?



Saito and Saito. TGANv2: Efficient Training of Large Models for Video Generation with Multiple Subsampling Layers. arXiv:1811.09245 (2018)

Want to learn more?



Clark et al. Adversarial Video Generation on Complex Datasets. arXiv:1907.06571 (2019)

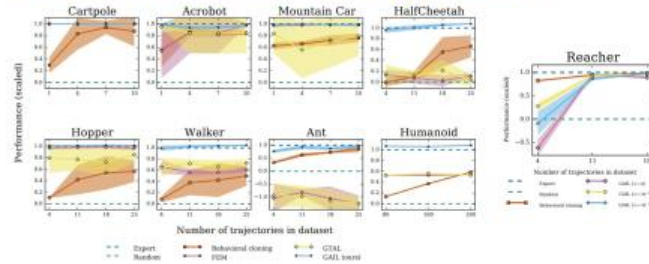
Want to learn more?



Luc et al. Transformation-based Adversarial Video Prediction on Large-Scale Data. arXiv:2003.04035 (2020)

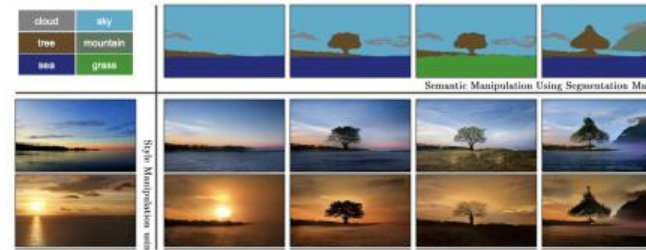
GANs Everywhere!

RL (Imitation Learning): **GAIL**



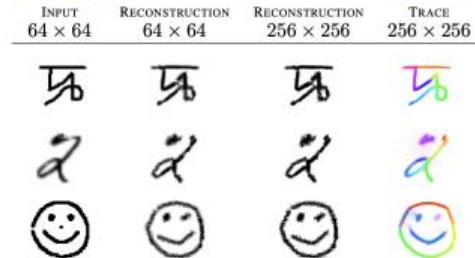
Ho and Erman. **Generative Adversarial Imitation Learning**. Neural Information Processing Systems (2016)

Image Editing: **GauGAN**



Park et al. **Semantic Image Synthesis with Spatially-Adaptive Normalization**. IEEE Conference on Computer Vision and Pattern Recognition (2019)

Program Synthesis: **SPIRAL**



Ganin et al. **Synthesizing Programs for Images using Reinforced Adversarial Learning**. International Conference on Machine Learning (2018)

Motion Transfer: **Everybody Dance Now**



Chan et al. **Everybody Dance Now**. International Conference on Computer Vision (2019)

Domain Adaptation: **DANN**



Ganin et al. **Domain-Adversarial Training of Neural Networks**. Journal of Machine Learning Research (2016)

Art: **Learning to See**



Akten. **Learning To See**. <http://www.memo.tv/portfolio/learning-to-see/> (2017, accessed 2020)

Evaluating generative models

Want to learn more?



Theis, et al A note on the evaluation of generative models
International Conference for Learning Representations (2016)

No evaluation metric is able to capture all desired properties.

- sample quality
- generalization
- representation learning

Evaluate based on end goal

- semi supervised learning: classification accuracy
- reinforcement learning: agent reward
- data generation: human (user) evaluation

Twenty-four quantitative techniques for evaluating GAN generator models

1. Average Log-likelihood

2. Coverage Metric

3. Inception Score (IS)

4. Modified Inception Score (m-IS)

5. Mode Score

6. AM Score

7. Frechet Inception Distance (FID)

8. Maximum Mean Discrepancy (MMD)

9. The Wasserstein Critic

10. Birthday Paradox Test

11. Classifier Two-sample Tests (C2ST)

12. Classification Performance

13. Boundary Distortion

14. Number of Statistically-Different Bins (NDB)

15. Image Retrieval Performance

16. Generative Adversarial Metric (GAM)

17. Tournament Win Rate and Skill Rating

18. Normalized Relative Discriminative Score (NRDS)

19. Adversarial Accuracy and Adversarial Divergence

20. Geometry Score

21. Reconstruction Error

22. Image Quality Measures (SSIM, PSNR and Sharpness Difference)

23. Low-level Image Statistics

24. Precision, Recall and F1 Score